MIL-T-44094A
24 October 1985
SUPERSEDING
MIL-T-44094
21 December 1982

MILITARY SPECIFICATION

TROUSERS, NIGHT CAMOUFLAGE, DESERT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers trousers made of cotton/nylon twill material.

1.2 Classification. The trousers shall be one type in the following sizes (see 6.2):

Schedule of sizes

Short	Regular	Long
X-small	X-small	X-small
Small	Small	Small
Medium	Medium	Medium
Large	Large	Large
X-large	X-large	X-large

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-280 - Thread, Gimp Cotton, Buttonhole
UU-P-268 - Paper, Kraft, Wrapping
DDD-L-20 - Labels; For Clothing, Equipage, and Tentage (General Use)

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-B-371 - Braid, Textile, Tubular
MIL-T-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-E-20652/1A - Eyelets, Metallic, Rolled Flange Type; and Eyelet Washer
MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic
MIL-T-43548 - Thread, Polyester, Cotton-Covered, and Rayon-Covered
MIL-C-44034 - Cloth, Twill, Camouflage Pattern, Cotton and Nylon for Desert Uniform

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1488 - Provisions for Evaluating Quality of Trousers

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Materials.

3.2.1 Basic cloth. The basic cloth shall conform to class 2 of MIL-C-44034.

3.2.1.1 Right fly lining, side opening facing and eyelet reinforcement pieces. The cloth for the right fly lining, side opening facing and eyelet reinforcement pieces shall be camouflage pattern basic material, ground shade cloth, or printed seconds cloth. Ground shade cloth shall be dyed in conformance with class 2 of MIL-C-44034 prior to over printing of the camouflage pattern. Ground shade cloth shall meet the physical, mechanical and dimensional requirements of the finished fabric. Printed seconds shall be cloth which has been rejected only for defects pertaining to color, infrared reflectance or camouflage pattern cited in class 2 of MIL-C-44034. If printing seconds are used, they must be marked on the face side with stripes, 3 inches apart across the full width running in a warpwise direction for the length of the entire roll. The stripes shall be 1/8 to 1/2 inch wide and shall be applied by the cloth finisher. The color of the stripes shall be black and have a colorfastness to crocking no lower than 6.5 (Munsell scale).

3.2.2 Braid, textile (tubular). The braid for waist drawcords shall be solid tubular cotton braid conforming to type IV, class 1, 4/32 inch diameter, Shade Olive Green 107 of MIL-B-371. The ends of the drawcords shall be dipped or impregnated with cellulose acetate or cellulose acetate butyrate, to prevent ravelling. The dipped or impregnated end shall be at least 1/2 inch in length.

3.2.3 Tape, textile, nylon. The tapes for the leg bottom drawcords shall be 3/8 inch wide, dyed Olive Green 106, conforming to type III, class 2 of MIL-T-5038.

3.2.4 Fastener tapes. The hook and pile fastener tapes shall be 1 inch wide for hip pockets, side openings and horizontal openings and 5/8 inch wide for fly closure, Olive Green 106, conforming to type II, class 1 of MIL-F-21840.

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3.2.5 Thread. The thread shall be polyester, cotton-covered or rayon-covered conforming to MIL-T-43548, ticket numbers 50, 2 or 3 ply and 70, 2 ply for all seaming and stitching and 50, 2 or 3 ply for buttonholes, eyelets and bar-tacks. The thread shall be dyed Olive Drab S-1, C.A. 66022.

3.2.5.1 Colorfastness. The thread shall show fastness to light and laundering (after 3 cycles), equal to or better than the standard sample (see 6.3). When no standard sample is available, the thread shall show "good" fastness to light and laundering (after 3 cycles).

3.2.6 Gimp. The cotton gimp for reinforcing buttonholes (when applicable) shall be size 8, Olive Drab S-1, C.A. 66022, conforming to type I or II of V-T-280.

3.2.6.1 Colorfastness. The gimp shall show fastness to laundering (after 3 cycles) equal to or better than the standard sample (see 6.3). When no standard sample is available, the gimp shall show "good" fastness to laundering (after 3 cycles).

3.2.7 Labels. Each trouser shall have an identification label, a size label and an instruction label conforming to type VI of DDD-L-20, except the label color shall be medium green, cable no. 70034, 70130 or 70131. The labels shall show fastness to laundering.

3.2.7.1 Identification label. The identification label shall conform to class 1.

3.2.7.2 Size label. The size label shall conform to class 2 and contain the following information for each respective size. Descriptive information shall be printed in letters and numerals not less than a minimum 10 point.

X-small-Short	X-small-Regular	X-small-Long
Waist: Below 27 in. Inseam: Up to 29-1/2 in. Stock No.	Waist: Below 27 in. Inseam: 29-1/2 to 32-1/2 in. Stock No.	Waist: Below 27 in. Inseam: Over 32-1/2 in. Stock No.
Small-Short	Small-Regular	Small-Long
Waist: From 27 to 31 in. Inseam: Up to 29-1/2 in. Stock No.	Waist: From 27 to 31 in. Inseam: 29-1/2 to 32-1/2 in. Stock No.	Waist: From 27 to 31 in. Inseam: Over 32-1/2 in. Stock No.
Medium-Short	Medium-Regular	Medium-Long
Waist: From 31 to 35 in. Inseam: Up to 29-1/2 in. Stock No.	Waist: From 31 to 35 in. Inseam: 29-1/2 to 32-1/2 in. Stock No.	Waist: From 31 to 35 in. Inseam: Over 32-1/2 in. Stock No.
Large-Short	Large-Regular	Large-Long

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Large-Short	Large-Regular	Large-Long
Waist: From 35 to 39 in. Inseam: Up to 29-1/2 in. Stock No.	Waist: From 35 to 39 in. Inseam: 29-1/2 to 32-1/2 in. Stock No.	Waist: From 35 to 39 in. Inseam: Over 32-1/2 in. Stock No.
X-large-Short	**X-large-Regular**	**X-large-Long**
Waist: From 39 to 43 in. Inseam: Up to 29-1/2 in. Stock No.	Waist: From 39 to 43 in. Inseam: 29-1/2 to 32-1/2 in. Stock No.	Waist: From 39 to 43 in. Inseam: Over 32-1/2 in. Stock No.

3.2.7.3 Instruction label. The instruction label shall conform to class 3, and contain the following information:

TROUSERS, NIGHT CAMOUFLAGE, DESERT

1. Wear as outer garment in desert areas for night camouflage.
2. Adjust waist by drawcord.
3. Wear leg bottoms outside footgear and secure drawcords
4. Wash by hand or machine; make sure all soap is thoroughly rinsed out.
5. DO NOT STARCH.

DO NOT REMOVE THIS LABEL

3.2.7.4 Combination identification and instruction label. The identification label (class 1) and instruction label (class 3) may be combined into one label. The combination label shall conform to the requirements of the class 3 label.

3.2.7.5 Combination size, identification and instruction label. The identification label (class 1), size label (class 2) and instruction label (class 3) may be combined into one label. The combination label shall conform to the requirements of the class 14 label. The coating requirement shall conform to class 3 label.

3.2.8 Eyelets. The metal eyelet for the waist drawcord outlet and bottom hem shall conform to dash No. BBE 124 of MIL-E-20652/1A except that the size shall be 1/4 ± 1/32 inch inside diameter.

3.3 Design. The trousers have a front opening, two side openings with flaps, two horizontal openings with flaps, two hip patch pockets with flaps (all have hook and pile fastener tape closures), waistband and leg bottom drawcords (see figure 1).

3.4 Patterns. Standard patterns for use in cutting working patterns will be furnished by the Government. The working patterns shall be identical to the Government patterns which shall not be altered in any way. The patterns provide for the following seam allowances, 1/2 inch for double lapped, double

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stitched seams and 3/8 inch for all other seams unless otherwise indicated. The patterns indicate size, placement of hand openings and notches for assembly of parts.

3.4.1 List of pattern parts. The component parts of the trousers shall be cut from the material specified and according to the number of parts listed in table I.

TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
Basic material		
Cotton/nylon twill cloth	Front	2
	Back	2
	Right fly lining	1
	Hip pocket	2
	Hip pocket flap	2
	Side opening flap	4
	Side opening facing	2
	Horizontal opening welts	4

3.5 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from the written document appear in figure 1, the written document shall govern.

3.5.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the trousers.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

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b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401, 502, and 503, stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a. and b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. Type 301 stitching may be used to repair 401 stitching.

3.5.2 Bartacks. The bartacks shall measure 3/8 to 1/2 inch in length.

3.5.3 Sew eyelets. The eyelets (when applicable) for trouser hems shall be clean-cut and the bight of stitching around the eyelet shall be not less than 1/16 inch and the inside diameter of the finished eyelet shall be 3/16 to 1/4 inch. The eyelets shall have the stitches overlapping not less than 1/16 inch. The stitches per eyelet shall be determined by counting the number of needle penetrations in the outside row of outside stitching. The overlapping stitches at the end of the stitching shall be included. When 401 stitch type is used, the purling shall finish on the outside of the trousers prior to hemming. When type 502 and 503 stitch type is used, the outside purling shall be at the point of needle penetration and the inside purling shall not be pulled to the outside. The inside purling shall finish on the outside of the trousers prior to hemming. The thread ends shall be inside the hem of the trousers.

3.5.4 Buttonholes. The buttonholes (when applicable) for the trouser hems shall be eyelet-end, taper-bar type worked over gimp. The bar end shall have a minimum of 4 tacks, not counting the first cross-over stitch. The purling shall finish on outside of trousers prior to hemming. The finished cut length of the buttonhole shall be 1/2 to 5/8 inch.

3.6 Manufacturing operations requirements. The trousers shall be made in accordance with the operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed, unless otherwise specified. Any additional basting or holding stitching used to facilitate manufacture is permissible provided the thread is removed or does not show on the finished trousers.

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TABLE II

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut material in strict accordance with patterns furnished, which show size, placement of pockets, directional lines, and notches for proper assembly. Directional lines indicate the warp direction of the material. The directional lines on the fronts and backs of the trousers parts may vary from the warp direction by not more than 2-1/2 inches. Measurements shall be taken from the top and bottom of the directional line on pattern to the selvage edge of the fabric and the difference between the two measurements shall not exceed 2-1/2 inches. b. Cut all parts from the same piece of material except right fly lining, side opening facings, side opening flap lining, metal eyelet reinforcement piece and buttonhole or eyelet reinforcement piece which may be cut from ends. c. Cut leg draw tapes (two per trouser) of sufficient length to allow for an extension of not less than 3 inches (unknotted) from each eyelet or buttonhole (see 3.2.3). d. Cut waistband drawcord (one per trouser) of sufficient length to allow for an extension of not not less than 9 inches (unknotted) from each exit (see 3.2.2). e. Cut the 1 inch hook and pile fastener tape for side openings, horizontal openings and hip pockets 1-1/8 ± 1/8 inches long.						

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NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) f. Cut the 5/8 inch hook and pile fastener tape for front opening in the following lengths for the sizes listed. 6-1/4: X-small-Short; X-small-Regular; X-small-Long; Small-Short; Small-Regular; Medium-Short 7-1/8: Small-Long; Medium-Regular; Medium-Long; Large-Short; Large-Regular 8: Large-Long; X-large-Short; X-large-Regular; X-large-Long g. Cut the reinforcement piece for metal eyelet, 1 inch by 1 inch, from the basic material or may be cut from printed seconds or ground shade material.						
2.	Component marking. a. Mark and ticket all components of the trousers except those parts cut from ends to insure correct size, proper assembly and a uniform shade throughout the trousers. b. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the trousers and the numbers are covered by the seam allowance.						

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Component marking. (cont'd)						
	c. Any method of marking may be used except:						
	1. Metal fastening devices. 2. Sew on type tickets. 3. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
3.	Replacement of defective components.						
	During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1488 shall be removed from production and replaced with non-defective and properly matched components.						
4.	Labeling.						
	a. Sew labels on all four sides, not more than 3/16 inch from edges, to trousers as follows:	301	LSbj-1	9-11	50	50	
	(1) Size label. Centered (1 inch off-center tolerance) on inside right back of trousers not more than 1/2 inch below the waistband stitching. The top edge of label may be caught in waistband stitching.						
	(2) Identification label. Centered (1 inch off-center tolerance) no more than 1 inch below the size label. The top edge of the identification label may be caught in the bottom of the size label stitching.						

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labeling. (cont'd)						
	(3) Instruction label. Centered (1 inch off-center tolerance) below the identification label. The stitching for the bottom edge of the identification label shall catch the top edge of the instruction label.						
	(4) Combination identification and instruction label. (see 3.2.7.4) Attach as specified in (2) above.						
	(5) Combination size, identification and instruction label. (see 3.2.7.5) Attach as specified in (1) above.						
5.	Make flaps for side hand openings.						
	a. Stitch the flap lining to flap along sides and pointed edges. Turn, work out edges and points, and stitch 1/16 to 1/8 inch from edges.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b)	9-11 9-11 9-11	50 50 50	50 70 50	
	b. Stitch 1 inch hook portion of fastener tape to underside of side hood opening flap lining as indicatedby marks on pattern.	301 or 401	LSbj-1 LSbj-1	9-11 9-11	50 50	50 70	
6.	Make hip pocket flaps.						
	a. Fold flap in half and seam sides. Turn, work out corners and edge stitch 1/16 to 1/8 inch from side edges and along folded edge.	301 or 401 and 301 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSf-1	9-11 9-11 9-11 9-11	50 50 50 50	50 70 50 50	
	or						

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make hip pocket flaps. (cont'd)						
	b. Fold flap in half, turning the side edges to the inside and stitch 1/16 to 1/8 inch from side edges and along folded edge.	301	SSc-1	9-11	50	50	
			and				
			OSf-1	9-11	50	50	
	or						
	c. Fold flap in half, turning all raw edges to the inside and stitch 1/16 to 1/8 inch from all edges.	301	SSc-1	9-11	50	50	
			and				
		301	OSf-1	9-11	50	50	
	d. Stitch 1 inch hook portion of fastener tape to underside of hip pocket flap on all sides as indicated by marks on patterns.	301 or	LSbj-1	9-11	50	50	
		401	LSbj-1	9-11	50	70	
7.	Make hip pockets.						
	a. Hem the top of pocket with the raw edges turned in and stitch 1/16 to 1/8 inch from turned edge.	301	EFb-1	9-11	50	50	
	b. Stitch 1 inch pile portion of fastener tape to hem, positioned as indicated by marks on pattern.	301	LSbj-1	9-11	50	50	
8.	Make horizontal opening welts.						
	a. Fold welts in half lengthwise and stitch sides and folded edge. (The stitching operation may be omitted.)	301	SSa-1	9-11	50	50	
	b. Stitch the 1 inch hook portion of fastener tape to top welt and pile portion to bottom welt, positioned as indicated by marks on pattern. Stitching shall be through both piles of the folded welt.	301	LSbj-1	9-11	50	50	

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Make and attach right fly lining.						
	a. Turn under top, bottom and back raw edges of right fly lining and stitch 1/16 to 1/8 inch from turned edge.	301	EFa-1	9-11	50	50	
	or						
	b. Overedge stitch top, bottom and back raw edges of right fly lining.	502 or 503	SSa-1 SSa-1	6-10 6-10	50 50	50 50	
	c. Stitch right fly lining to front of trouser, positioned as indicated by pattern. Turn right fly lining to inside and stitch 1/16 to 1/8 inch from edge.	301 or 401 and 301	SSa-2(a) SSe-2(a) SSe-2(b)	9-11 9-11 9-11	50 50 50	50 70 50	
	d. Stitch 5/8 inch pile portion of fastener tape, on all four sides, to outside of right front, with stitching through fly lining. The length of the fastener tape shall be centered on finished fly opening (1/8 inch off center tolerance permitted) and the front edge of fastener tape shall be 1 inch (+0, -1/8) from finished edge of fly.	301	LSbj-1	9-11	50	50	
	or						
	e. If fastener tape is stitched to sides with double needle machine, top and bottom ends of tape may be bartacked with a horizontal bartack. Bartack shall be positioned no more than 1/8 inch from top and bottom ends of tape.	Bartack		28 per bartack	50	50	
10.	Make left fly.						
	a. Turn under raw edges of left fly and stitch 1/16 to 1/8 inch from turned edge.	301	EFa-1	9-11	50	50	
	or						
	b. Overedge stitch raw edges of left fly.	502 or 503	SSa-1 SSa-1	6-10 6-10	50 50	50 50	

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MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make left fly. (cont'd)						
	c. Fold left fly to inside of trousers at marks indicated on pattern and stitch 1/16 to 1/8 inch from folded edge.	301	OSf-1	9-11	50	50	
	d. Stitch 5/8 inch hook portion of fastener tape to inside of left front, with the front edge of tape 1/8 ± 1/16 inch from finished front edge of fly and the length centered on the finished fly opening (1/8 inch) off center tolerance is allowed.	301	LSbj-1	9-11	50	50	
	or						
	Bartack top and bottom ends of tape if sides are stitched with double needle machine	Bartack		28 per bartack	50	50	
11.	Set facings and flaps for side hand openings.						
	a. Seam side hand opening facing to each front, from notch to notch, with a 5/8 inch seam allowance. Turn and stitch on edge of facing 1/16 to 1/8 inch.	301 or 401 and 301	SSe-2(a) SSE-2(a) SSe-2(b)	9-11 9-11 9-11	50 50 50	50 50 50	
	b. Turn under the back raw edge of facing and stitch 1/16 to 1/8 inch from edge.	301	LSd-1	9-11	50	50	
	or						
	Overedge stitch top, back and bottom raw edge of facing, and stitch through front 1/8 inch from overedge stitch.	502 or 503	EFd-1	6-10	50	50	
	c. Position flaps and stitch to hold on each front in proportion above and below seam cut opening on 5/8 inch seam allowance.	301	SSa-1	9-11	50	50	
	d. Position and stitch a one inch pile of fasteners tape to correspond to back portion of flap.	301	LSbj-1	9-11	50	50	

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MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Sew on hip pockets and flaps.						
	a. Position pockets as indicated by marks on pattern, turn in the sides and bottom of pockets and stitch to trousers 1/16 to 1/8 inch from turned edges.	301	LSd-1	9-11	50	50	
	b. Seam top edge of flap to trousers above pocket opening 3/16 inch from edge, as indicated by marks on patterns. Turn and stitch 3/16 to 1/4 inch from turned edges.	301	LSbk-2	9-11	50	50	
	or						
	c. When all raw edges of hip pocket flaps are turned in (see oper. 6.b. or c), position as indicated by marks on pattern and stitch 3/16 to 1/4 inch from edge.	301	LSbj-1	9-11	50	50	
13.	Join outseams and form side hand opening.						
	Join outseams with a double lapped and double stitched seam, with the backs of trousers lapping fronts, catching the flaps and knee pleat in the joining seam. The hand openings shall be formed by not catching the front in the seaming under the flap. The outside folded edge of pleats shall face toward bottom of trousers.	301 or 401	LSc-2 LSc-2	9-11 9-11	50 50	50 70	
14.	Join crotch seam.						
	a. Turn crotch seam of right front to the outside and stitch 1/16 to 1/8 inch from edge.	301 or 401	LSdc-3(a) LSdc-3(a)	9-11 9-11	50 50	50 70	
	b. Turn edge of left front crotch seam to the inside overlap on right front and stitch with two rows of stitching 3/16 to 1/4 inch apart catching the right crotch seam with both rows of stitching.	301 or 401	LSdc-3(b) LSdc-3(b)	9-11 9-11	50 50	50 70	

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Finish left fly.						
	Stitch back edge of fly, 1/16 to 1/8 inch from edge. Stitching shall be from top of trouser and curve to fly notch.	301	LSbj-1	9-11	50	50	
16.	Make horizontal openings on outseam.						
	a. Position welts and stitch as indicated by marks on pattern (marks indicate finished dimensions of welts). Cut openings and tongue notch to ends of stitching.	301 or 401	LSbj-1 LSbj-1	9-11 9-11	50 50	50 70	
	b. Turn welts to inside (top lapped on bottom) and with side edges of opening turned in, raise stitch the opening on all four sides 1/16 to 1/8 inch from edge.	301	LSb-1	9-11	50	50	
17.	Join seat seam.						
	Join seat seam with a double lapped, double stitched seam.	401	LSc-2	9-11	50	70	
18.	Join inseams.						
	Join inseams with a double lapped, double stitched seam with backs or trousers lapping fronts, forming knee pleat in the joining seam. The outside folded edge of the pleats shall face toward the bottom of trousers. The crotch seam and seat seam may be staggered the width of the seam.	301 or 401	LSc-2 LSc-2	9-11 9-11	50 50	50 70	

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Attach eyelet and hem waist.						
	a. Position eyelet reinforcement on inside of right front on marks for eyelet, and attach eyelet through reinforcement piece and trousers. The eyelet shall be centered (3/16 inch off center tolerance) on the reinforcement piece.						
	b. Turn top of trousers over drawcord at notches provided on pattern and with raw edge turned in, stitch with two rows of stitching. The top row shall be 1/8 to 1/4 inch from top edge and the bottom row shall be 1/16 to 1/8 inch from bottom turned edge with stitching continued down right front edge the width of the waistband. The ends of drawcord shall exit through the eyelet on the right front and at the front of waistband on the left. The drawcord shall not be caught in the stitching of waist hem.	301 or 401	EFt-2	9-11	50	50	
	c. Place two vertical bartacks on the left front waist hem above and below drawcord. The bartacks shall be 1/8 to 1/4 inch from front edge. The drawcord shall not be caught in the bartack stitching. On left front drawcord shall exit approximately in the center of the waistband.	Bartack		28 per bartack	50	50	
	d. The ends of the drawcord shall be knotted to prevent return into hem.						

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
20.	Hem bottom of trousers.						
	a. Make two buttonholes to finish in center of underside of trousers hem through reinforcement piece at front of each trouser leg, as indicated on pattern. Position the eyelet end of each buttonhole toward front of trouser. Reinforcement piece shall be of sufficient length and width to cover the area surrounding the outlets. The reinforcement pieces may be pre-stitched into position prior to making drawcord outlets.	Buttonhole		44-46 per buttonhole	50	50	
	or						
	b. Make two sewn eyelets through reinforcement, positioned as indicated for the eyelet end of the buttonhole.	Eyelet		19-24 per eyelet	50	50	
	or						
	c. Make two 1/2 to 5/8 inch, straight cut buttonholes tacked at both ends, through reinforcement and positioned as indicated on pattern.	301		44-46 per buttonhole including tacks	50	50	
	or						
	d. Attach metal eyelets through reinforcement piece and trouser as indicated on pattern. The eyelets shall be centered (3/16 off-center tolerance) on the reinforcement piece and positioned as indicated in operation 20 b.						
	e. Thread drawcords through outlets of trousers.						
	f. Turn up bottom of trousers over cord, as indicated by marks on pattern, turn under raw edge and stitch 1/16 to 1/8 inch from turned edge. The drawcords shall not be caught in the stitching of trouser hems. The hem shall finish 1 ± 1/4 inches wide.	301	EFb-1	9-11	50	50	

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TABLE II (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Bartacking.						
	a. Bartack both top corners of each hip pocket with vertical bartack positioned on pocket stitching.	Bartack		28 per bartack	50	50	
	b. Bartack the base of fly with a diagonal bartack positioned on the fly stitching.	Bartack		28 per bartack	50	50	
	c. Bartack the top and bottom end of each side hand opening flap with a horizontal bartack through flap, trousers, and facing pieces. The bartacks shall be positioned 1/4 to 3/8 inch from ends of flap.	Bartack		28 per	50	50	
	d. Bartack each leg drawcord through hem at center of back (± 1/2 inch off-center tolerance) with a vertical bartack.	Bartack		28 per	50	50	
22.	Cleaning.						
	a. Trim all ends of stitching and remove loose threads.						
	b. Remove all spots and stains.						
	c. Remove all size tickets.						

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3.7 Measuarements. The finished trousers shall conform to measurements shown in table III.

TABLE III. Finished measurements (in inches)

Size	1/2 Waist 1/	Inseam 2/ Short	Regular	Long
X-small	15-3/4	30-1/2	32-1/2	34-1/2
Small	17-3/4	30-1/2	32-1/2	34-1/2
Medium	19-3/4	30-1/2	32-1/2	34-1/2
Large	21-3/4	30-1/2	32-1/2	34-1/2
X-large	23-3/4	30-1/2	32-1/2	34-1/2
Tolerance	± 1/2	± 3/4	± 3/4	± 3/4

1/ Measurement taken along top edge of waist from folded edge to folded edge with fly closed.

2/ Measurement taken form center of crotch seam to bottom of leg.

3.8 Workmanship. The finished trousers shall conform to the quality of product established by this document. The occurence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of all inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When a certificate of compliance is submitted, the Government reserves the right to check test such items and determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, or

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qualified in this document or applicable purchase documents. The contractor shall furnish a certificate of compliance stating that the dipping or impregnating compound conforms to the requirements in 3.2.2.

4.2.2 Point count inspection. Sampling and inspection provisions for end item examination of utility (u) trousers, dimensional examination, and packaging inspection shall be performed in accordance with MIL-STD-1488.

4.2.3 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defects
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each pair of trousers, with inseams meeting, shall be folded from the leg bottom up toward the waistband to an overall length of approximately 23 inches. For the larger sizes it may be necessary to adjust the seat area to achieve the desired width of approximately 14-1/2 inches. Ten pairs of folded trousers of one size only shall be neatly stacked in a bundle, with the waistbands alternated, and securely tied with cotton tape or twine approximately four inches from each end.

5.1.2 Commercial. Trousers shall be preserved in accordance with ASTM D-3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Forty pairs of trousers of one type and size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A intermediate packs shall be packed flat, four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B packing. Forty pairs of trousers of one type and size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat, four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.

5.2.3 Commercial packing. Trousers, preserved as specified in 5.1, shall be packed in accordance with ASTM D-3951.

5.3 Palletization. When specified (see 6.2) trousers, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry) type III, type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means

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K or L or film bonding means O and P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs (bundles) and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

5.4.1 Labels, mixed sizes. Each shipping container, packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSN's" plainly stamped or printed thereon and under these words shall be legibly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The trousers are a component of the desert combat uniform and are intended for wear by military personnel of the Department of the Army to provide nighttime camouflage in desert areas.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Size required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(e) When palletization is required (see 5.3).

6.3 Sample. For access to sample of the end item, address the contracting activity issuing the invitation for bids.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8415-0454

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA-CT

User activity:

Air Force - 45

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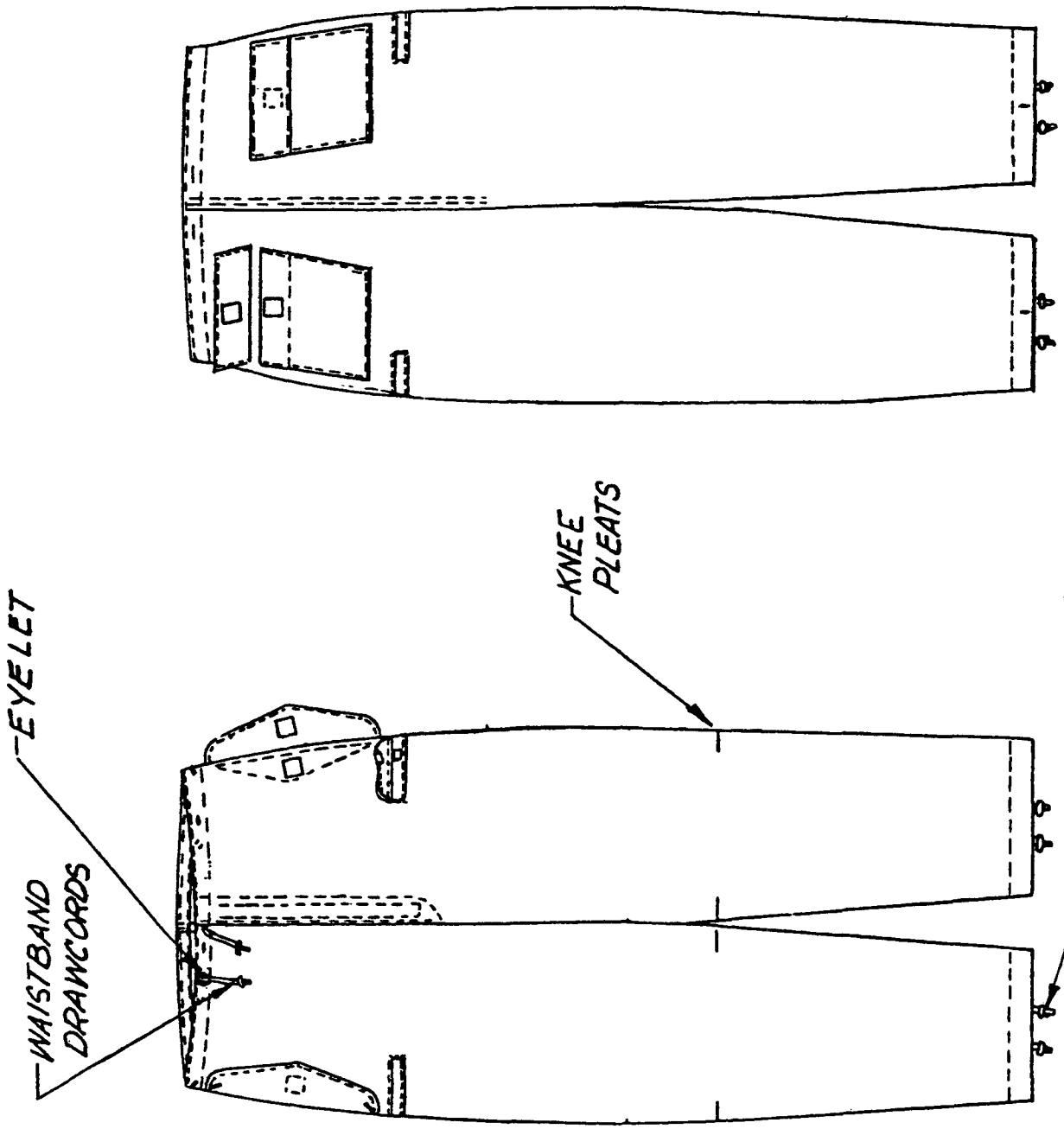


FIGURE 1. TROUSERS, NIGHT CAMOUFLAGE, DESERT

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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and DEVELOPMENT CENTER
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Natick, MA 01760-5014

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL–T–44094A	TROUSERS, NIGHT CAMOUFLAGE, DESERT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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